

Executive Brief: Sales Forecasting & Stocking Strategy

A Data-Driven Audit and 3-Month Demand Plan for the CFO & Head of Supply Chain

To: Chief Financial Officer & Head of Supply Chain

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Subject: Multi-Source Sales
Forecasting & Inventory
Optimization

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Live Dashboard: sales-forecasting-and-anomaly-detection.streamlit.app

1. Executive Summary

Getting inventory replenishment levels right determines the profitability of our retail operations. Overstocking locks up working capital in warehouse storage, whereas understocking directly translates to lost sales and customer churn. This report delivers a rigorous demand intelligence audit by training and comparing three statistical and machine learning models on 4 years of daily transactional data. Our findings show that our sales exhibit strong seasonality, peaking aggressively in November and December. By deploying the recommended **SARIMA forecasting model** (which achieved the lowest test error with an MAE of \$19,299.92), we have generated a concrete 3-month outlook. Furthermore, we leverage weekly anomaly detection and product-level K-Means clustering to provide a precise stocking strategy per demand segment.

2. Key Exploratory Findings

- **Revenue Anchor:** The **Technology** category represents our highest revenue-generating division, pulling in the largest absolute sales volume. Furniture and Office Supplies represent steady secondary lines.
- **Consistent Growth Vector:** The **West Region** demonstrates the most consistent sales growth over the 4-year period, showing steady growth rates year-over-year. Other regions exhibit higher volatility in growth.
- **Shipping Lag:** The average time elapsed between Order Date and Ship Date stands at **3.96 days**. This duration is highly consistent and shows virtually no statistical variation across regions (ranging narrowly from 3.92 to 4.06 days).
- **Peak Seasonality:** An aggressive sales spike consistently occurs every November and December, representing a seasonal increase of **45-60%** compared to Q1-Q2 monthly baselines.
- **Multi-Source Correlation:** Merging our retail data with regional Video Game Sales data reveals that our Technology category sales are highly correlated with broader electronics/entertainment cycles (corr: 0.88), indicating that retail inventory should align with regional electronics launch calendars.

3. 3-Month Sales Forecast (Jan - Mar 2019)

Based on evaluation metrics, the **SARIMA model** was chosen for production. The model projects total sales of **\$161,508.96** over the next 3 months, broken down as follows:

Month	Forecasted Sales (\$)	Lower Bound (95% CI)	Upper Bound (95% CI)
Month 1 (Jan 2019)	\$51,910.46	\$45,681.20	\$58,139.72
Month 2 (Feb 2019)	\$37,726.17	\$32,444.51	\$43,007.83
Month 3 (Mar 2019)	\$71,872.33	\$61,091.48	\$82,653.18

Total Quarter 1	\$161,508.96	\$139,217.19	\$183,800.73
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4. Weekly Sales Anomalies & Root Causes

Using Isolation Forest and Rolling Z-Score audits, we scanned our weekly sales to detect statistically unusual behaviors. Top anomalies detected include:

- 1. Week Ending March 22, 2015 (\$37,703.67):** Sales deviated over 2.5 standard deviations from the rolling mean. Likely cause is *End of quarter bulk purchasing or distributor deals*, requiring dynamic inventory adjustments.
- 2. Week Ending December 02, 2018 (\$35,998.90):** Sales deviated over 2.5 standard deviations from the rolling mean. Likely cause is *Pre-holiday restocking and corporate promotional sales*, requiring dynamic inventory adjustments.
- 3. Week Ending November 18, 2018 (\$30,572.45):** Sales deviated over 2.5 standard deviations from the rolling mean. Likely cause is *Pre-holiday restocking and corporate promotional sales*, requiring dynamic inventory adjustments.

5. Product Demand Segmentation (K-Means)

Applying K-Means clustering to product sub-categories reveals four distinct demand profiles. replenishing items according to these profiles will maximize our capital efficiency:

Segment Type	Sub-Categories	Stocking Replenishment Strategy
1. High Value, Volatile (Copiers, Machines)	Copiers, Machines	Implement Just-In-Time (JIT) contracts. Keep safety stock low to avoid tying up capital in high-cost, slow-moving items.
2. Cash Cows (Stable) (Chairs, Phones)	Chairs, Phones, Storage, Binders	Maintain high safety stock (>95% service level) . Order in bulk to minimize shipping fees and capture vendor discounts.
3. Growing / Emerging (Accessories, Paper)	Accessories, Paper, Art	Implement monthly rolling review . Gradually scale up inventory capacity as demand indicates growth.
4. Declining / Low Volume (Supplies, Fasteners)	Supplies, Fasteners, Envelopes	Keep minimal stock levels . Bundle slow-moving items with high-margin items to liquidate warehouse space.

6. Data-Backed Business Recommendations

- 1. Automate Pre-Holiday Inventory Buildup:** Implement a mandatory inventory increase of 45% for the 'Cash Cows' segment starting October 15th annually. This locks in shipping capacity and prevents stockouts during the high-volume holiday period.
- 2. Transition to JIT replenishment for Copiers and Machines:** Establish direct drop-shipping or pre-negotiated fast shipping contracts with manufacturers. This mitigates overstock risk and frees up warehouse floor space for high-volume items.
- 3. Synchronize Stocking with Tech Release Cycles:** With a strong correlation (0.88) between general technology sales and gaming industry releases, the marketing and purchasing teams should coordinate stock levels based on Q3/Q4 electronic product release calendars.

7. Forecast System Risk & Limitations

Model Blindspots to External Shocks: Our models are trained on historical sales patterns and assume that the future will mirror past trend and seasonal cycles. They cannot anticipate black-swan events, such as supply chain disruptions, macroeconomic shifts, or direct competitor actions. We advise reviewing and re-fitting the models monthly, combining the forecast outputs with qualitative market intelligence from regional managers.